Shoeb Sheikh(CS23074)

Practical 9 Extra

```
# Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score
```

```
from google.colab import drive
drive.mount("/content/drive")
```

```
Mounted at /content/drive
```

```
path= "/content/drive/MyDrive/ML_Dataset/WineQT.csv"
df=pd.read_csv(path)
df.head(5)
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality	Id
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	5	0
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.68	9.8	5	1
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.65	9.8	5	2
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58	9.8	6	3

```
X = df.iloc[:, [3, 4]].values
```

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

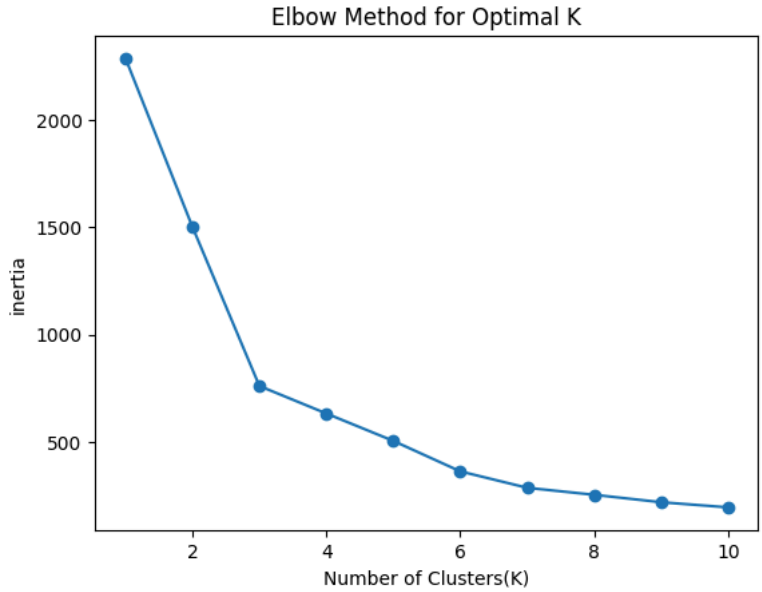
```
inertia_values = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    inertia_values.append(kmeans.inertia_)

print(inertia_values)

plt.figure()
plt.plot(range(1,11), inertia_values, marker='o')
plt.title('Elbow Method for Optimal K')
plt.xlabel('Number of Clusters(K)')
plt.ylabel('inertia')
plt.show()
```

```
[2286.000000000005, 1501.3598860914428, 761.3961622476105, 633.4432682350536, 505.9773376469779, 364.2213841246995, 287.7285
```




```
kmeans = KMeans(n_clusters=6, random_state=42)
clusters = kmeans.fit_predict(X_scaled)

# Add cluster labels to dataset
df['Cluster'] = clusters
(df.head())
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality	Id	Cluster
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	5	0	4
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.68	9.8	5	1	5
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.65	9.8	5	2	5
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58	9.8	6	3	4

```
print("Inertia:",kmeans.inertia_)

sil_score = silhouette_score(X_scaled, clusters)
print("Silhouette Score:", sil_score)
```

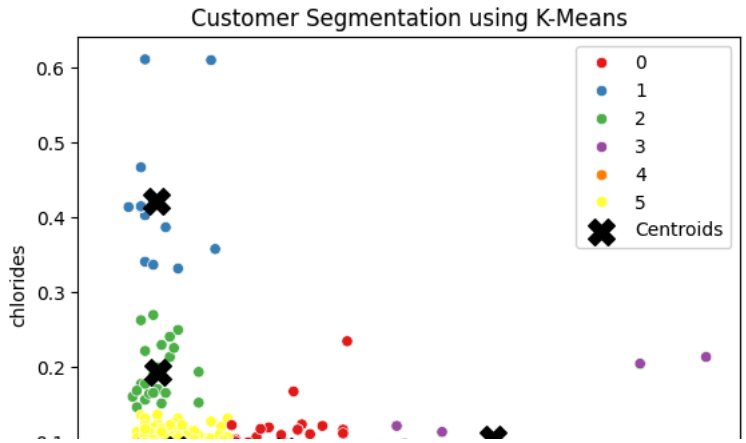
```
Inertia: 364.2213841246995
Silhouette Score: 0.375760646202804
```

```
plt.figure()
sns.scatterplot(x='residual sugar',
                y='chlorides' , hue='Cluster',
                data=df, palette='Set1'
)

centroids = scaler.inverse_transform(kmeans.cluster_centers_)

plt.scatter(
    centroids[:, 0],
    centroids[:, 1],
    s=200, c='black', marker='X',
    label='Centroids'
 )

plt.title("Customer Segmentation using K-Means")
plt.legend()
plt.show()
```


Customer Segmentation using K-Means
chlorides
0.6
0.5
0.4
0.3
0.2
0
1
2
3
4
5
Centroids